

Seattle Transit Frequency Dashboard

Main Goals

The main goals for this project are to create a dashboard to display kcmetro and sound transit routes and be able to filter the display by frequency at different parts of the day. The ability to filter time more generally and allow routes from other services will come, but the main goal is to create a barebones dashboard that is useable by the others.

Workflow

To start, I will have to download and unzip the kcmetro and sound transit data. From then I will take the stop times, group by route_id, stop_id, time frame, direction, sort by departure time.

Calculate median gap, average gap, max gap, min gap, bottom 25%, top 25%,

Save data by route with various information.

Dashboard to display route shapes – use arcgis or folium

Dashboard filter to display only certain routes

Walkshed Bubbles

Eventually time to certain destinations

Create .py files that create gap data and save it, create dashboard

Files Needed

Data:	Note:
Kcmetro transit data	For raw data to transform
Sound transit data	Raw data to transform
gap_calc.py	Create and save gap data by route_id
Display.py	Map display in pdf or interactable form
Info.py	Create pyplot figures about info of various routes

Information to Calculate

Frequency data by route – various metrics
Route shapes
Most frequent routes, least frequent routes, etc pyplot

First and last trips, running hours, speed, time to downtown, etc
Realtime information to detect delays and actual service headways with traffic